MIL-P-47005(MI)
19 April 1974
SUPERSEDING
MIS-14210
1 March 1965

MILITARY SPECIFICATION

PASTE, COLOR CONCENTRATE, EPOXY RESIN BASE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of organic paste color concentrates for use as colorants of epoxy resin systems.

1.2 Classification. The color concentrates covered by this specification shall be of the following types, as specified (see 6.2):

Type I: Blue (phthalocyanine blue).

Type II: Black (carbon black).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Federal

FED-STD-141 Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing

FSC 5970

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Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 305	Acetone Extract in Black Pigments
ASTM D 1652	Epoxy Content of Epoxy Resins
ASTM D 1726	Hydrolyzable Chlorine Content of Liquid Epoxy Resins

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample (see 4.2) of the color concentrate shall meet the requirements of this specification.

3.2 Material. The color concentrates shall consist of finely divided organic pigments, phthalocyanine blue and carbon black (see 1.2), thoroughly milled into a vehicle of pure epoxy liquid resin. The epoxy resin shall be a type that is 100 percent reactive.

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3.3 Chemical properties. The color concentrates shall meet the chemical properties requirements specified in Table I.

Table I

Chemical Properties

Property	Value		Applicable Test
	Type I	Type II	
Pigment content, percent, by weight	22.5 to 23.5	9.5 to 10.5	4.6.1.1
Epoxy assay, gram per gram mole	174 to 186	174 to 186	4.6.1.2
Hydrolyzable chlorine, percent, maximum	0.30	0.30	4.6.1.3

3.4 Physical properties.

3.4.1 Weight. The weight in pounds per gallon for Type I and Type II color concentrate shall be 10.4 plus or minus 0.2 and 10.5 plus or minus 0.2 pounds respectively.

3.4.2 Condition in container. A freshly opened full container shall show no curdling, livering, caking, lumps, skins, or excessive settling. There shall be no evidence of dirt, foreign material or other contaminants. The color concentrates shall mix readily to a smooth homogeneous state.

3.4.3 Shelf life. The shelf life of the color concentrates in unopened containers shall be a minimum of one year from date of manufacture when stored at temperature less than 27 degrees Celsius (C) (80.6 degrees Fahrenheit (F)).

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3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of material. The preproduction sample shall be subjected to all examinations and tests specified herein. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Classification of examinations and tests. The examination and testing of material shall be classified as follows:

a. Preproduction tests (see 4.4).

b. Quality conformance tests (see 4.5).

4.4 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.5 Quality conformance tests. Quality conformance tests for acceptance of material shall consist of the following examinations and tests:

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a. Pigment content (see 4.6.1.1).

b. Condition in container (see 4.6.2.2).

4.5.1 Lot size and sampling.

4.5.1.1 Lot size. Lot size shall consist of all the color concentrate manufactured at one time from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.5.1.2 Samping. Unless otherwise specified (see 6.2), 1/2 pint from one unit, representative of each lot (see 4.5.1.1), shall be selected at random for quality conformance testing. Each container in lot shall be considered as a unit of product. Failure of sample to meet the quality conformance test requirements shall be cause for lot rejection.

4.6 Test methods.

4.6.1 Chemical property tests.

4.6.1.1 Pigment content. The pigment content specified in Table I shall be determined in accordance with ASTM D 305.

4.6.1.2 Epoxy assay. The epoxy assay specified in Table I shall be determined in accordance with ASTM D 1652.

4.6.1.3 Hydrolyzable chlorine. The hydrolyzable chlorine content specified in Table I shall be determined in accordance with ASTM D 1726.

4.6.2 Physical property tests.

4.6.2.1 Weight. The weight per gallon specified in 3.4.1 shall be determined in accordance with FED-STD-141, Method 4184.

4.6.2.2 Condition in container. The condition of the material in container specified in 3.4.2 shall be determined visually and by stirring the contents of the container and examining for uniformity of dispersion and caking. Caking is defined as material which cannot be readily dispersed into the material by stirring.

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4.6.2.3 Shelf life. The material shall be tested for conformance with all the requirements specified herein after completion of the storage period. The manufacturer's certificate of compliance with the shelf life requirement (see 3.4.3) may be accepted during the storage period.

4.7 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be examined for compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the procuring activity, the material may be furnished in tin plate cans or pails with metal lids. The containers shall be non-reactive to the material. Size of the container shall be specified by the procuring activity. The material shall be packaged in such a manner that damage during shipment and storage is prevented.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The paste color concentrates covered by this specification are intended for use as colorants for epoxy resin encapsulating systems. The degree of color brightness or opacity required will determine amount of colorant used in the mixture of epoxy resin system (see 6.3).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1).

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c. Sampling plan if other than specified (see 4.5.1.2).

d. Any special marking requirements (see 5.2).

e. Type of color concentrate (see 1.2).

f. Container size (see 5.1).

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14210, dated 1 March 1965.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 5970-A467

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